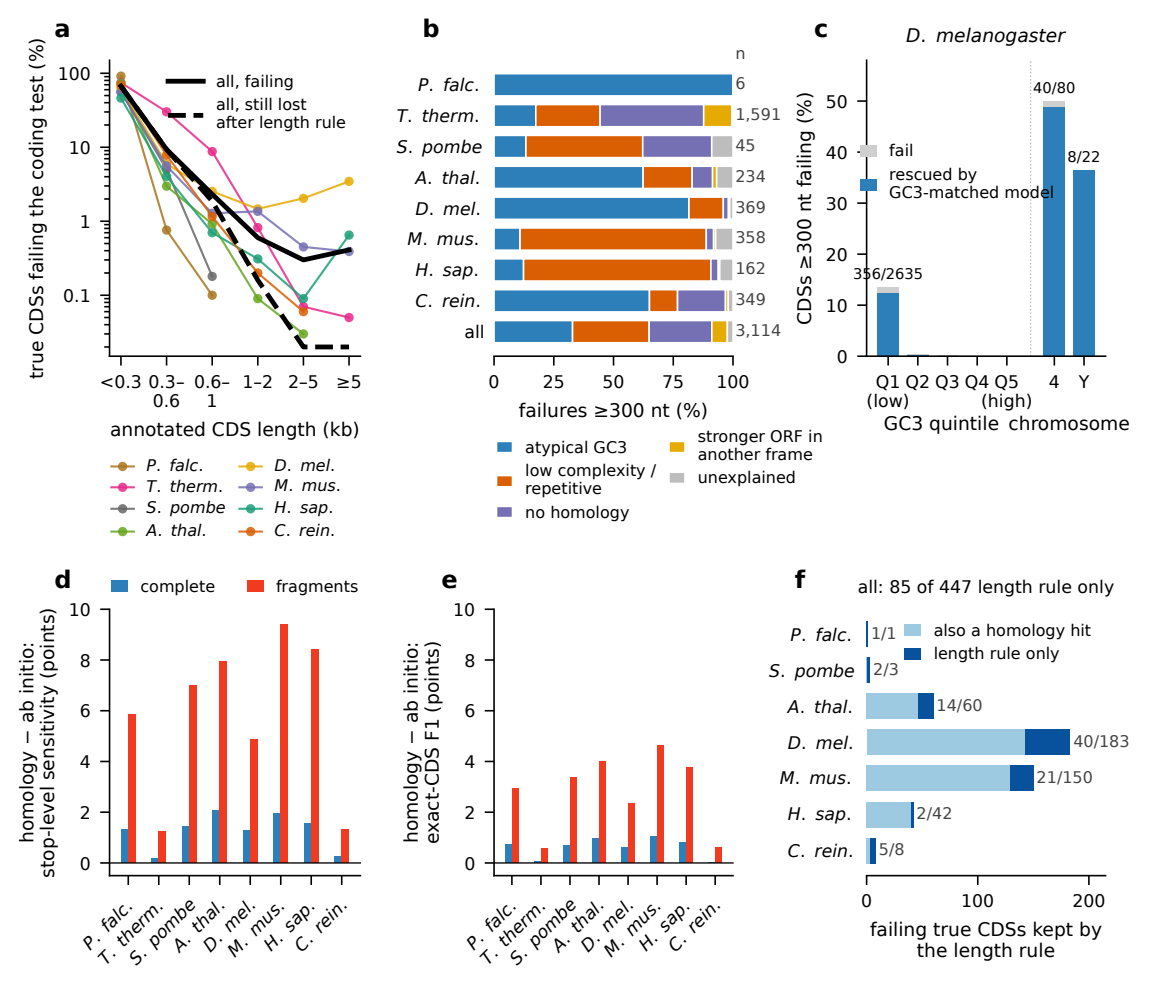
a
true CDSs failing the coding test (%)
100
10
1
0.1
all, failing
all, still lost after length rule
<0.3
0.3–0.6
0.6–1
1–2
2–5
≥5
annotated CDS length (kb)
P. falc.
T. therm.
S. pombe
A. thal.
D. mel.
M. mus.
H. sap.
C. rein.
b
n
P. falc. 6
T. therm. 1,591
S. pombe 45
A. thal. 234
D. mel. 369
M. mus. 358
H. sap. 162
C. rein. 349
all 3,114
0
25
50
75
100
failures ≥300 nt (%)
atypical GC3
low complexity / repetitive
no homology
stronger ORF in another frame
unexplained
c
D. melanogaster
CDSs ≥300 nt failing (%)
50
40
30
20
10
0
fail
rescued by GC3-matched model
356/2635
40/80
8/22
Q1 (low)
Q2
Q3
Q4
Q5 (high)
4
Y
GC3 quintile
chromosome
d
complete
fragments
homology − ab initio: stop-level sensitivity (points)
10
8
6
4
2
0
P. falc.
T. therm.
S. pombe
A. thal.
D. mel.
M. mus.
H. sap.
C. rein.
e
homology − ab initio: exact-CDS F1 (points)
10
8
6
4
2
0
P. falc.
T. therm.
S. pombe
A. thal.
D. mel.
M. mus.
H. sap.
C. rein.
f
all: 85 of 447 length rule only
also a homology hit
length rule only
P. falc. 1/1
S. pombe 2/3
A. thal. 14/60
D. mel. 40/183
M. mus. 21/150
H. sap. 2/42
C. rein. 5/8
0
100
200
failing true CDSs kept by the length rule